BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

INTERMEDIATE (12TH CLASS) EXAMINATION, 2025

MATHEMATICS

Past Paper 2025

Retyped faithfully from the original board paper — same questions, no changes.

UNSOLVED, Mathematics, Inter Part-II, 2025, Lahore Board, Group - II, Time: 2.30 hrs., Essay Type, Marks : 80, SECTION - I

Time Allowed: 2.30 hrs. | **Maximum Marks:** 80

Note: SECTION – I

Q2. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) Show that the parametric equations $x = a \sec \theta$, $y = b \tan \theta$ represent the equation of hyperbola $x^2/a^2 - y^2/b^2 = 1$

(ii) Without finding the inverse, state the domain and range of f^{-1}, where $f(x) = 2 + \sqrt{(x - 1)}$

(iii) Define continuity of a function at a number C.

(iv) Evaluate $\lim_{\theta \to 0} (1 - \cos \theta)/\sin \theta$

(v) Find fog(x) and gof(x), when $f(x) = 2x + 1$ and $g(x) = 3/(x - 1)$, $x \neq 1$

(vi) Find dy/dx, if $x = at^2$ and $y = 2at$

(vii) Prove that $d/dx (\tan^{-1} x) = 1/(1 + x^2)$, $\forall x \in R$

(viii) If $x = y \sin y$, find dy/dx

(ix) Find y_2, if $y = \ln ((2x + 3)/(3x + 2))$

(x) Apply the Maclaurin Series expansion to prove that $\cos x = 1 - x^2/2! + x^4/4! - x^6/6! + - - -$

(xi) Determine the intervals in which f is increasing or decreasing, $f(x) = \cos x$; $x \in (-\pi/2, \pi/2)$

(xii) Differentiate $x^2 - 1/x^2$ w.r.t x^4

Q3. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) Find δy of the function $y = x^2 - 1$ when x changes from 3 to 3.02.

(ii) Evaluate $\int (\sin \theta \, d\theta)/(1 + \cos^2 \theta)$

(iii) Find $\int x \sin x \, dx$

(iv) Evaluate the integral $\int dx/(x \ln x)$

(v) Find the definite integral $\int (x \, dx)/(x^2 + 2)$ from 1 to e^(tan⁻¹ x)

(vi) Evaluate $\int$ e^(tan⁻¹ x) $dx/(1 + x^2)$

(vii) Solve the differential equation $(x - 1)dx + y \, dy = 0$

(viii) Define inclination of a line.

(ix) Check whether the lines $4x - 3y - 8 = 0$, $3x - 4y - 6 = 0$ and $x - y - 2 = 0$ are concurrent.

(x) Find an equation of line through A (-6, 5) having slope 7.

(xi) Convert $15y - 8x + 3 = 0$ into normal form of line.

(xii) Define homogeneous function.

Q4. (Marks 18) Write short answers to any NINE (9) questions:

(i) Write the procedure for graphing a linear inequality in two variables.

(ii) Draw a graph of linear inequality $x + 2y < 6$

(iii) Show that the equation $5x^2 + 5y^2 + 24x + 36y + 10 = 0$ represents a circle. Also find its centre and radius.

(iv) Check the position of point (5, 6) with respect to circle $2x^2 + 2y^2 + 12x - 8y + 1 = 0$.

(v) Write an equation of parabola with focus (1, 2) and vertex (3, 2).

(vi) Find foci and eccentricity of ellipse $9x^2 + y^2 = 18$

(vii) Find the centre and directrix of hyperbola $x^2/4 - y^2/9 = 1$

(viii) Find a unit vector in the direction of the vector $y = 2i + 6j$

(ix) Find the vector from the point A to the origin where $AB = 4i - 2j$ and B is the point (-2, 5)

(x) Find a and b so that the vectors $3i - j + 4k$ and $ai + bj - 2k$ are parallel.

(xi) Find the projection of a along b when $a = 3i + j - k$, $b = 2i - j + k$

(xii) Find a vector perpendicular to vectors $a = 2i + j + k$, $b = 4i + 2j - k$

(xiii) Differentiate between scalar quantity and vector quantity.

> Note: SECTION – II

> Note: Note: Attempt any THREE questions.

Q5.

(a) [5] If f(x) = { 3x if $x \leq -2$; $x^2 - 1$ if $-2 < x < 2$; 3 if $x \geq 2$ — Discuss continuity at x = 2 and x = -2

(b) [5] Differentiate $(x^2 + 1)/(x^2 - 1)$ w.r.t. $(x - 1)/(x + 1)$

Q6.

(a) [5] Find the extreme values of the function defined by $f(x) = x^3 - 6x^2 + 9x$

(b) [5] Evaluate the integral $\int (5x + 8)/((x + 3)(2x - 1))\, dx$

Q7.

(a) [5] Evaluate $\int$ from 0 to $\pi/4$ $\cos^4 t\, dt$

(b) [5] Maximize f(x,y) = 2x + 3y subject to the constraints $2x + y \leq 8$, $x + 2y \leq 14$, $x \geq 0$, $y \geq 0$

Q8.

(a) [5] Find a joint equations of the lines through the origin and perpendicular to the lines $x^2 - 2xy \tan \alpha - y^2 = 0$

(b) [5] Find an equation of circle which passes through the points A (5,10), B (6, 9) and C (-2, 3)

Q9.

(a) [5] If a = 4i + 3j + k and b = 2i - j + 2k, find a unit vector perpendicular to both a and b. Also find the sine of angle between them.

(b) [5] Show that an equation of the parabola with focus at (a cos α, a sin α) and directrix x cos α + y sin α + a = 0 is $(x \sin \alpha - y \cos \alpha)^2 = 4a(x \cos \alpha + y \sin \alpha)$.